WASP-39b

R-1854

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-39_260603` · created 2026-08-03T08:37:17+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461194.6993 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.012
Transit depth (Rp/Rs)^2	3.35 +/- 0.44 [%]
Orbital Inclination (inc)	89.97 +/- 1.02
Airmass coefficient 1 (a1)	1.269 +/- 0.023
Airmass coefficient 2 (a2)	-0.158 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.79 %
Best Comparison Star	None
Optimal Aperture	4.37
Optimal Annulus	16.38
Transit Duration (day)	0.0696 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.012 vs 0.1457 (deviation 2.26σ)
Duration fit vs published	0.0696 d vs 0.12 d (deviation 7.81σ)
Mid-time O–C	-57.2 min
Depth SNR	11.1
Residual scatter	1.79 %

Light curve

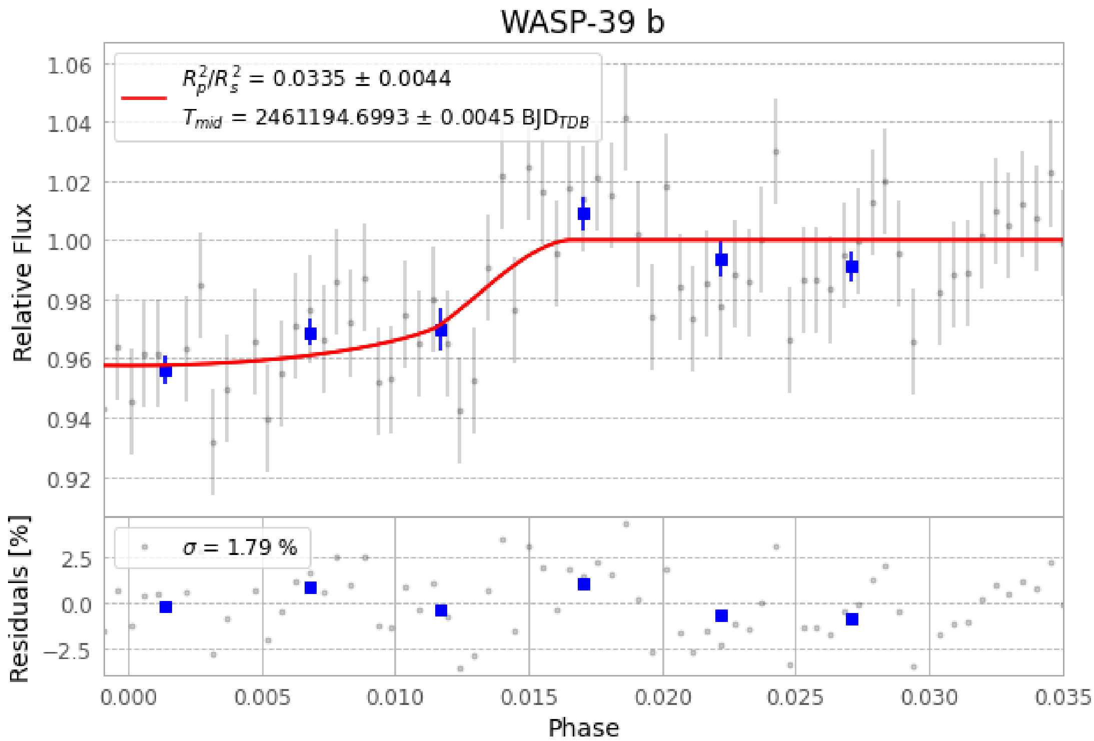

Figure 1 — FinalLightCurve_WASP-39 b_2026-06-02.png (SHA-256 967712395e129c58…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 153], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c48aefcf645cc1d0…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

70 FITS frames (46 MB), `WASP-39260603043315.FITS` → `WASP-39260603080315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-260603.log (7f7940f90ae0…), FinalLightCurve_WASP-39 b_2026-06-02.png (967712395e12…), FinalLightCurve_WASP-39 b_2026-06-02.csv (c5f30bf798b2…)

Compute resources

Wall time 175.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)